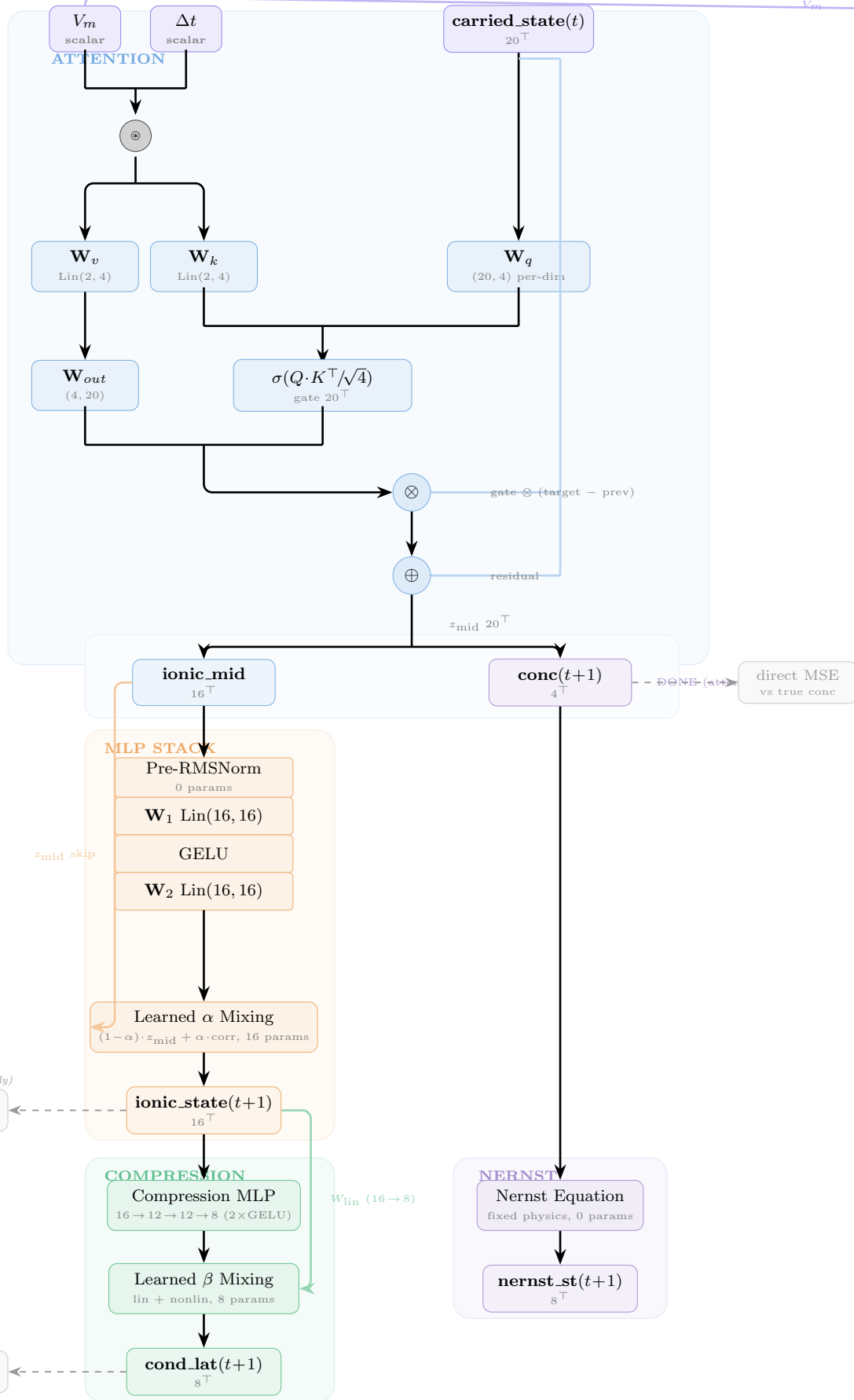


Ionic Surrogate v3

1,454 inference params · 970 critical-path FLOPs · 3,292 background FLOPs

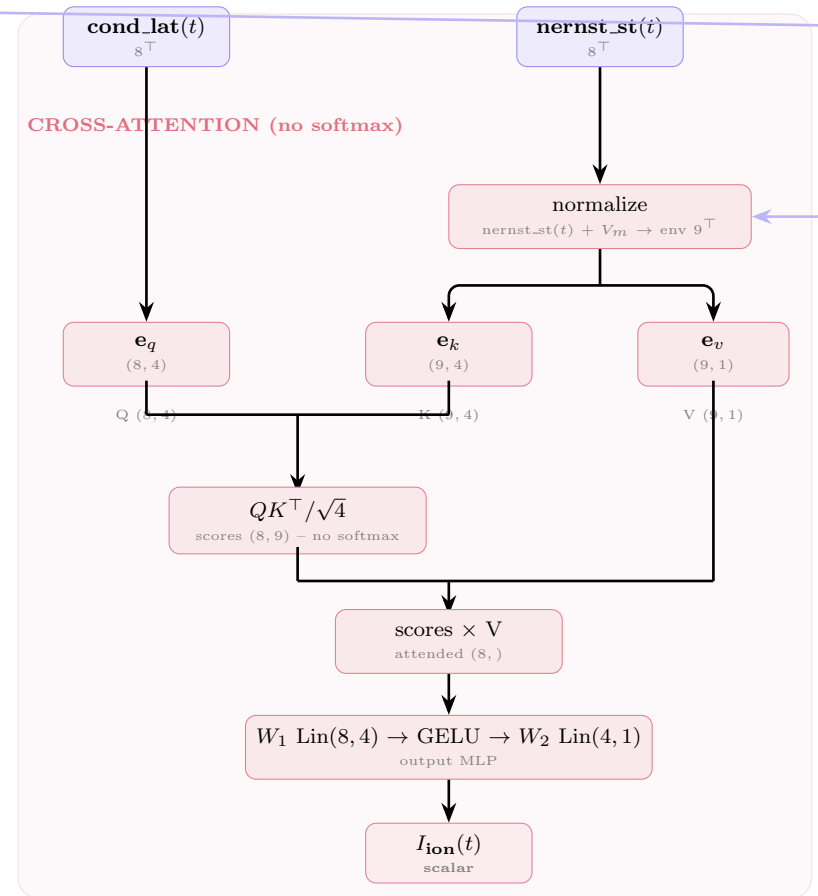
Stage 1: State Evolution

off critical path · 1,336 params



Stage 2: Current Readout

ON critical path · 118 params



External: $V_m^{n+1} = V_m + \Delta t(-I_{ion} + I_{stim})/C_m$ — I_{stim} applied in operator splitting, not a model input. Stage 1 runs on separate CUDA stream during diffusion solve.

Legend:



elem. add

gate $\otimes$ (target - prev)

stack

data flow (1.0pt)

training only (0.7pt)

merge/split bar